



Mark Scheme

Mock Set 4

Pearson Edexcel GCE
In AS Level Mathematics (8MA0)
Paper 21 Statistics

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

EDEXCEL GCE MATHEMATICS

General Instructions for Marking

1. The total number of marks for the paper is 30.
2. The Edexcel Mathematics mark schemes use the following types of marks:
 - **M** marks: method marks are awarded for 'knowing a method and attempting to apply it', unless otherwise indicated.
 - **A** marks: Accuracy marks can only be awarded if the relevant method (M) marks have been earned.
 - **B** marks are unconditional accuracy marks (independent of M marks)
 - Marks should not be subdivided.
3. Abbreviations
These are some of the traditional marking abbreviations that will appear in the mark schemes.
 - bod – benefit of doubt
 - ft – follow through
 - the symbol $\surd$ will be used for correct ft
 - cao – correct answer only
 - cso - correct solution only. There must be no errors in this part of the question to obtain this mark
 - isw – ignore subsequent working
 - awrt – answers which round to
 - SC: special case
 - oe – or equivalent (and appropriate)
 - dep – dependent
 - indep – independent
 - dp decimal places
 - sf significant figures
 - * The answer is printed on the paper
 - $\square$ The second mark is dependent on gaining the first mark
4. For misreading which does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, in that part of the question affected.
5. Where a candidate has made multiple responses and indicates which response they wish to submit, examiners should mark this response.
If there are several attempts at a question which have not been crossed out, examiners should mark the final answer which is the answer that is the most complete.
6. Ignore wrong working or incorrect statements following a correct answer.
7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternatives answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

Question	Scheme	Marks	AOs
1(a)	J and L	B1	1.2
		(1)	
(b)	$0.27 + v + 0.2 + u + 0.18 = 1$ $u + v = 0.35 \quad *$	B1*	1.1b
		(1)	
(c)(i)	Use of independence $P(J) \times P(K) = P(J \text{ and } K)$ and substituting to get an equation in terms of u and v to reach $(0.27 + u)(u + v + 0.2) = u$	M1	2.1
	Substitution of $u + v = 0.35$ to form an equation in u only $0.55(u + 0.27) = u \quad \text{o.e.}$	A1	1.1b
		(2)	
(c)(ii)	Solving their equation from (i) to find a value of u $0.55(u + 0.27) = u$ $0.45u = 0.1485$ $u = \dots$	M1	1.1b
	$u = 0.33$ $v = 0.02$	A1	1.1b
		(2)	
(6 marks)			
Notes:			
(a)	B1:	Correctly choosing J and L only	
(b)	B1*:	Writing or using sum of probabilities = 1 and simplifying to reach $u + v = 0.35$	
(c)(i)	M1:	Use of independence relationship to obtain an equation in u and v	
	A1:	Substituting into independence equation to reach an equation in u only	
(c)(ii)	M1:	Solving their equation to obtain a value for u	
	A1:	Correct values of u and v	

Question	Scheme		Marks	AOs
2(a)	Use of $X \sim B(40, 0.29)$		M1	1.1b
	$P(X \geq 11) \quad [= 1 - P(X \leq 10)] = 0.64104\dots$	0.641 *	A1*	2.1
			(2)	
(b)(i)	$Y \sim B(5, 0.641\dots)$		M1	3.3
	$P(Y \leq 1) \quad [= P(Y = 0) + P(Y = 1)]$		M1	3.4
	awrt 0.0592		A1	1.1b
			(3)	
(b)(ii)	The proportion of customers choosing meat-free meals each day is constant or Each evening is independent		B1	3.5b
			(1)	
(c)	$H_0: p = 0.29 \qquad H_1: p \neq 0.29$		B1	2.5
	$X \sim B(40, 0.29)$		M1	3.3
	$P(X \leq 6) = 0.03219$ 0.0322	awrt	A1	1.1b
	[0.0322 > 0.025, accept H_0] Insufficient evidence to suggest the proportion of customers choosing menu B has changed.		A1	2.2b
			(4)	
(10 marks)				
Notes:				
(a)	M1:	Use of the correct distribution, may be implied by sight of e.g. 0.359, 0.764 or 0.236		
	A1*:	$P(X \geq 11)$ and 0.641 or better, with no incorrect working		
(b)(i)	M1:	Writing or using the correct distribution and $P(X \geq 11)$		
	M1:	Use of $P(Y \leq 1)$, might see $P(Y = 0) + P(Y = 1)$ $[(1 - 0.641\dots)^5 + 5(0.641\dots)(1 - 0.641\dots)^4]$ Can be implied by correct answer		
	A1:	awrt 0.0592		
(b)(ii)	B1:	Correct assumption in context		
(c)	B1:	For both hypotheses correct and in p or π		
	M1:	Writing the correct distribution, or evidence of using it		
	A1:	awrt 0.0322		
	A1:	Correct statement including proportion, customers, and vegan/meat-free/ menu B Do not allow contradictory statements		

Question	Scheme		Marks	AOs
3(a)(i)	(Order the data by date and) use simple random sampling to select the first piece of data		B1	1.2
	$\frac{184}{30} = 6.13...$ Select the first data point then every 6 th piece of data until 30 pieces of data have been collected		B1	2.2a
			(2)	
(a)(ii)	Using the same dates as for Leeming		B1	2.4
			(1)	
(b)	Camborne is nearer to the coast than Leeming and The mean cloud cover in Camborne is greater than that of Leeming		M1	2.4
	... this does not support Joe's belief		A1	2.2b
			(2)	
(c)	(Discrete) uniform distribution		B1	3.3
	Assuming that each <u>okta</u> of cloud cover is equally likely		B1	3.5b
			(2)	
(7 marks)				
Notes:				
(a)(i)	B1:	Describing using a random method to find the first piece of data/starting point. Condone no mention of ordering.		
	B1:	Describing taking every 6 th piece of data until 30 pieces of data are collected (allow use of a total number of days in the range 178-186 inclusive)		
(a)(ii)	B1:	For noting that the same dates selected for Leeming should be used		
(b)	M1:	Indicating that Camborne is nearer to the coast than Leeming, as well as mentioning that Camborne's mean is greater		
	A1:	Correct reasoning and a statement that this does not support Joe's belief		
(c)	B1:	Uniform distribution or discrete uniform distribution B0 for continuous uniform distribution		
	B1:	Indicating that all outcomes are equally likely – must use context by mentioning 'oktas'		
Allow the following as an alternative for (c)				
(c) ALT	B1:	Allow B(8, p)		
	B1:	e.g. assuming cloud cover for each okta is independent		

Question	Scheme		Marks	AOs
4(a)	$\sigma_x^2 = \frac{110}{20} - \left(\frac{32}{20}\right)^2$		M1	1.1b
	$\sigma_x^2 = 2.94$		A1	1.1b
			(2)	
(b)	$3.5 \times 14 (= 49)$		B1	1.1b
	$\frac{32 + 3.5 \times 14}{20 + 14} (= 2.38\dots)$		M1	2.1
	$2.5 = \frac{\Sigma y^2}{14} - 3.5^2$ $\Sigma y^2 = 206.5$		M1	2.1
	Combined variance = $= \frac{110 + '206.5'}{20 + 14} - ('2.38\dots')^2$		M1	1.1b
	awrt 3.63		A1	1.1b
			(5)	
(7 marks)				
Notes:				
(a)	M1:	Correct equation for variance		
	A1:	2.94 or equivalent		
(b)	B1:	For finding or using Σy		
	M1:	Correct method for finding combined mean seen or used		
	M1:	$\Sigma y^2 = 206.5$ stated or used		
	M1:	Correct method for combined variance, using their Σy^2 and their combined mean		
	A1:	awrt 3.63		